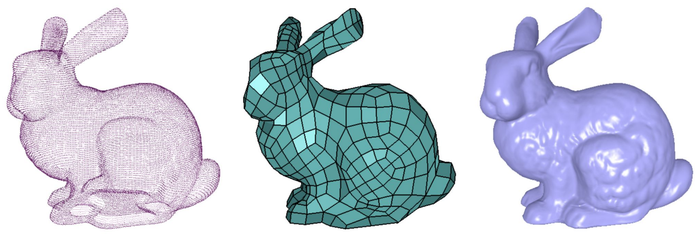
Convolutional Occupancy Networks

A. Chaikova, O. Dabooni, K. Gelvan, Y. Wang

3D Object Reconstruction

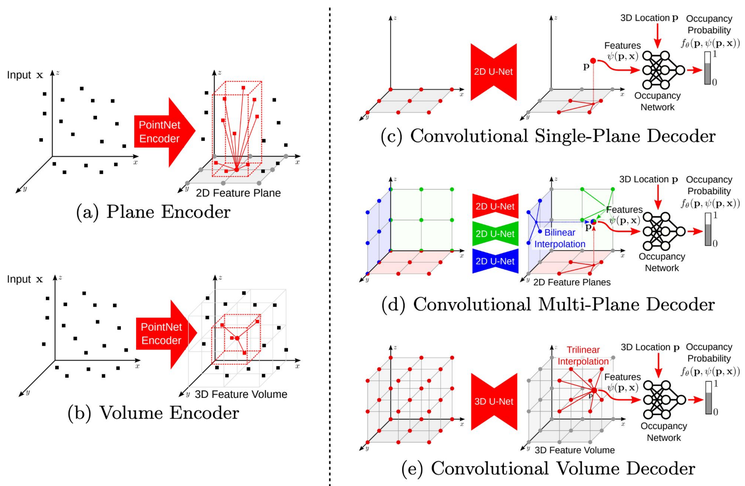


(a) (b) (c)

(a) point cloud, (b) control mesh, (c) result

- **Input:** point clouds (table, car, airplane + bus, bathtub)
- **Output:** 3D mesh
- **Importance:** Used in imaging, navigation, design
- **Challenges:** Complex shapes, thin structures, noise

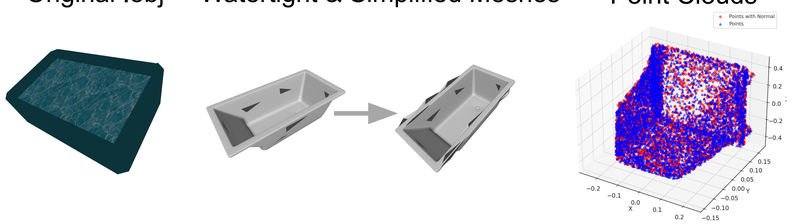
Convolutional Occupancy Networks



- Transform raw 3D data to a 2D feature plane
- Represent features on a normal grid
- Concatenate query points with feature vector
- Classify with Occupancy Net (after bilinear interpolation)

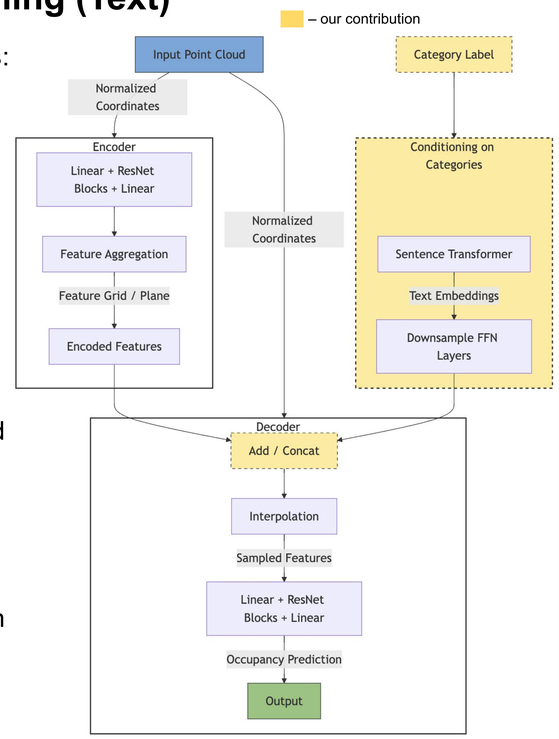
Data pre-processing for bus & bathtub classes

Original .obj | Watertight & Simplified Meshes | Point Clouds



Auxiliary conditioning (Text)

- **Baseline Embeddings**: Embedding layer (3- or 32-dim)
- **Pretrained model:** *MiniLM-v2* (384-dim)
- **Combination method**: Add or Concat
- Further training yields better results compared to using frozen model
- Embeddings concatenation outperforms summation



Positional encoding

- Uses sin/cos encoding from NeRF
- 10 frequency bands => 20 values for each coordinate
  - ➢ Best when applied on the query 3D point, before the Occupancy Net

Table 1: Positional Encodings variants

Configuration	IoU
Baseline (3plane_3class)	0.798
+PosEnc(Decoder)	0.820
+PosEnc(Decoder)+Norm	**0.824**
+PosEnc(2x)	0.608
+PosEnc(2x)+Norm	0.527

(2x) is PosEnc added both before the encoder & decoder.

Different activation functions in reconstruction

- Use of periodic (sinusoidal) activation function with and without PosEnc
  - ➢ Brings 5% quality improvement!

Table 2: Activation Function variants

Configuration	IoU
Baseline (3plane_3class)	0.798
+Sin	**0.837**
+Sin+PosEnc(Decoder)	0.816
+Sin+PosEnc(2x)	0.510

(2x) is PosEnc added both before the encoder & decoder.

Results

- Additional conditioning on category
  - ➢ Leads to an **increase** in quality
- Positional encoding
  - ➢ Decoder-only PosEnc **outperform** Encoder-Decoder ones
- Sinusoidal activation functions
  - ➢ Does indeed **improve** the baseline

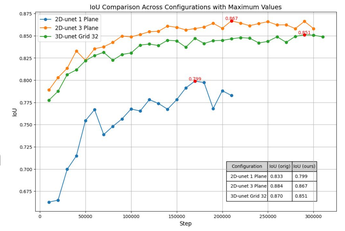


Figure 1: Comparison of the baseline trainings

Table 3: Comparison of training on 3 most popular classes (IoU)

Configuration	Add	Concat
Baseline (3plane_3class)	0.798	
Pretrained Encoder* (frozen)	**0.798**	0.794
Pretrained Encoder* (unfrozen)	0.787	**0.809**
Simple Encoder	0.793	**0.810**

*MiniLMv2 with down-projection

Conclusions

- Trained over **20** models
- **All** the conducted experiments for 3D reconstruction **improve the overall quality** compared to the baseline
- Most improvement is brought by the **sinusoidal activation function**



Future Work

- Conduct more experiments to calculate the IoU variance
- Fix data generation pipeline
- Combine the successful ideas and try them on other 3D tasks

Code:

*gray 3D visualizations are done by trained models on validation examples